
Problem 8.21. For any positive integer x, let x^R be the integer whose binary representation is the reverse of the binary representation of x (Assume no leading 0s in the binary representation of x.) Define the function $R^+ : \mathbb{N} \longrightarrow \mathbb{N}$ where $R^+(x) = x + x^R$.

Part a. Let $A_2 = \{\langle x, y\rangle \mid R^+(x) = y\}$. Show $A_2 \in \mathrm{L}$.

Proof. To show that $A_2 \in \mathrm{L}$, we give a log space algorithm M that decides A_2.

$M =$ "On input $\langle x, y\rangle$, where x and y are positive binary integers without leading 0s:

1. Let i, j, n_x, n_y be binary integers, and s and c be binary digits on work tape.
2. Let $n_x = |x|$, and $n_y = |y|$.
3. Let $i = 1$, $j = n_x$, and $c = 0$.
4. Repeat for n_x steps:
5. If y does not have i^{th} digit ($i > n_y$), then *reject*.
6. Let s be the sum of ith and j^{th} binary digits of x and c. If the addition produces a carry, then $c = 1$. Otherwise, $c = 0$.
7. If s and i^{th} digit of y are different, then *reject*.
8. Increment i by 1, and decrement j by 1.
9. If $c = 1$, then:
10. If y does not have i^{th} digit, then *reject*.
11. If c and i^{th} digit of y are different, then *reject*.
12. If y has $(i + 1)^{th}$ digit, then *reject*.
13. Otherwise, if y has i^{th} digit, then *reject*.
14. *Accept*."

□

Part b. Let $A_3 = \{\langle x, y\rangle \mid R^+(R^+(x)) = y\}$. Show $A_3 \in$ L.

Proof. A log space algorithm that decides A_3 can be devised like follows:

1. Compute first and last binary digits of $R^+(x)$ using the algorithm presented in Part a.
2. Add the binary digits and keep track of any carry for later use.
3. Compare the least significant digit of the sum with the least significant digit of y. *Reject*, if they are different.
4. Next, iteratively compute the second and second last digits, third and third last digits and so forth.
5. Take sum of the two digits computed in each step and any carry from previous step, and compare the least significant digit of the sum with the corresponding digit in y. *Reject*, if the digits differ.
6. Finally, perform checks related to final carry and extra digits in y.

□